

Project Title:

Analysing Commuting Trends and Challenges in Melbourne's Rail Network

Introduction

Melbourne's rail network is a critical component of the city's public transport system, serving millions of commuters annually. Recent debates about overcrowding and service reliability have dominated local news, with reports highlighting peak-hour congestion at major stations like Flinders Street and Southern Cross (Victoria State Government, 2023). Understanding commuting patterns and their influencing factors is essential for urban planners, transport authorities, and policymakers to optimize service efficiency, reduce overcrowding, and improve passenger safety. This project aims to explore how train usage varies across time, quantify station crowding, and assess the impact of weather on ridership.

Motivation

As a frequent user of Melbourne's rail network, I have experienced firsthand the challenges of overcrowding during peak hours and delays caused by extreme weather. My background in urban studies has fueled my interest in leveraging data to address public transport inefficiencies. By analyzing ridership patterns and external factors like weather, I hope to contribute actionable insights that enhance commuter experiences and support sustainable urban mobility.

Questions

1. How do train service frequency and passenger numbers fluctuate across different time periods, and what are the internal and external factors (e.g., weather conditions) that influence these variations?
2. What methods can be used to quantify station crowding? How do factors (e.g., time of day, station design, service frequency) contribute to station crowding, and which stations are most affected?

Data Sources

A. Train Service Passenger Counts (FY July 2023–June 2024)

This tabular dataset (CSV format) contains estimated passenger boarding and alighting counts for Melbourne's rail services. With 15.6 million rows and 21 columns, it includes attributes such as *Business Date*, *Day Type* (e.g., weekday, public holiday), *Line Name*, *Direction* (toward/away from Flinders Street), and aggregated passenger counts (e.g., boardings, alightings). Spatial attributes like station coordinates and temporal granularity (hourly intervals) enable analysis of ridership trends and station-specific patterns. This dataset will be used to address Questions 1 and 2. URL: [Train Service Passenger Counts](#).

B. Argyle Square Weather Stations Historical Data (May 2021–July 2024)

This tabular dataset (CSV format) includes 138,538 rows of hourly weather observations. Variables such as *precipitation*, *windspeed*, *air temperature*, and *relative humidity* will be merged with ridership data to assess weather impacts. Combined with Data Source A, this dataset will help answer Question 3. URL: [Argyle Square Weather Data](#).

C. Public Transport Stops (November 2024)

This spatial dataset (GeoJSON format) provides geographic coordinates and metadata for 28,323 public transport stops across Victoria, including Melbourne's rail network. Attributes include *Stop ID*, *Stop Name*, and *Mode* (e.g., Metro Train, Regional Train). It supports mapping station locations and integrating spatial context with Data Source A for visualisation. URL: [Public Transport Stops](#).

D. Public Transport Lines (November 2024)

Another spatial dataset (GeoJSON format) details 9,776 public transport routes, including rail lines. Attributes like *Line Name*, *Headsign* (destination), and *Mode* facilitate analysis of service coverage and connectivity. It complements Data Source C to contextualize passenger flows across the network spatially. URL: [Public Transport Lines](#).

References

Victoria State Government. (2023). *Melbourne's rail network performance report*. Public Transport Victoria. <https://www.ptv.vic.gov.au/performance>